



Mark Scheme (Results)

June 2024

Pearson BTEC Nationals in
Applied Science/Forensic and Criminal
Investigation (31617H/1P)
Unit 1: Principles and Applications of Science I
(Physics)

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Unit 1: Principles and Applications of Science I (Physics)

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Additional Guidance	Mark
1(a)(i)	2.0 (cm)	accept 2 accept two	1
1(a)(ii)	30 (cm)	accept thirty	1
1(b)	C 180°		1
1(c)	<u>substitution (1)</u> (wave speed =) $3.5 \div 2.2$ <u>evaluation (1)</u> 1.6(ms ⁻¹)	answer 1.6(ms ⁻¹) without working gains full marks accept 1.59(ms ⁻¹) accept correct rounding of 1.590909091 to at least 2 significant figures	2
Total = 5 marks			

Question Number	Answer	Additional Guidance	Mark
2(a)	compression(s) (1) rarefaction(s) (1)	answers in either order	2
2(b)	bat {emits/gives out/ produces} {sound/ultrasound/signal/pulse/wave} (1) (reflection/echo from the prey) travels back to bat (1) time (delay between emission and receipt of signal) gives distance from prey (1)	accept makes accept noise ignore references to prey emitting sound/bat wings producing sound ignore high pitch/ frequency accept bounces back do not accept volume gives distance from prey	3
Total = 5 marks			

Question Number	Answer	Additional Guidance	Mark
3(a)(i)	B star 2		1
3(a)(ii)	identification hydrogen (1) expansion lines in hydrogen (spectrum) are present in all (star) spectra (1)	any element other than hydrogen scores 0 marks	2
3(b)	Explanation including: (before heating) each element has a different set of electron energy levels (1) AND any two from: <u>electrons</u> fall/drop to a lower energy level (1) energy (change) emitted as {light/photon/e-m wave} (1) the change in energy levels is equal to the energy of the photon emitted (1)	accept different configuration accept electrons de-excite accept emit light of a specific frequency/wavelength	3
Total = 6 marks			

Question Number	Answer	Additional Guidance	Mark
4(a)	C from glass into air/is greater than the critical angle		1
4(b)	<u>substitution (1)</u> $\frac{3(.0) \times 10^8}{v} = \frac{\sin 40}{\sin 25}$ <u>rearrangement (1)</u> $(v =) \frac{3(.0) \times 10^8 \times \sin 25}{\sin 40}$ <u>evaluation of sine (1)</u> $(v = \frac{3(.0) \times 10^8 \times 0.4226}{0.6428})$ <u>evaluation of v (1)</u> $2.0 \times 10^8 (\text{m s}^{-1})$	answer $2(.0) \times 10^8 (\text{m s}^{-1})$ without working gains full marks $(v = 3(.0) \times 10^8 \times 0.6574)$ accept 2×10^8 and 1.97×10^8 accept 197 000 000 evaluation of v to any other power of ten gains 3 marks accept correct rounding (with working) of evaluation of v to at least 2 significant figures	4
Total = 5 marks			

Question Number	Answer	Additional Guidance	Mark
5(a)	Any one from (many) changes in frequency (each second) (1) frequency/channel hopping (1)	do not accept different (frequency/wavelength/wave)	1
5(b)	Any two differences from: infrared has higher frequency/shorter/lower wavelength (1) infrared needs 'line of sight'/ does not pass through/is absorbed by obstacles (1) infrared affected / interference by bright sunlight (1)	or reverse arguments for Bluetooth® accept different frequency/wavelength do not accept infrared has lower frequency/higher wavelength (or reverse incorrect statements for Bluetooth®) ignore uses/applications ignore how many devices can be connected	2

Question number	Indicative content	
5(c)	<i>Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and levels descriptors below. The indicative content that follows is not prescriptive. Answers may cover some or all of the indicative content, but learners should be rewarded for other relevant answers.</i> • mobile phones are used on a system of networks • no network coverage by particular supplier • network (e.g. 3G switched off by provider) • base stations transmit and receive signals over a limited distance • out of range of base station (transmitter) • network too busy • (mobile phone) providers are allocated a band of frequencies in the radio/microwave region • limited bandwidth available • two transmitters giving destructive interference • being underground/in a tunnel • signals blocked by walls/buildings/objects • interference by weather • (e.g. rain, moisture) absorbs microwaves	
Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates adequate knowledge of scientific facts/concepts with generalised comments made. • Generic statements may be presented rather than linkages made so that lines of reasoning are unsupported or partially supported. • The discussion shows some structure and coherence.
Level 2	3–4	• Demonstrates good knowledge and understanding by selecting and applying some relevant scientific facts/concepts to provide the discussion being presented. • Lines of argument mostly supported through the application of relevant evidence. • The discussion shows a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of scientific facts/concepts to provide the discussion being presented. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The discussion shows a well-developed structure which is clear logical and coherent.

Total = 9 marks



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